OCEAN IMPRINTS
Bioplastic × Saltwater

Bioplastics are often presented as a sustainable alternative to conventional plastics. Yet, there is not one bioplastic, but many recipes, each with different properties and behaviors.

So how do we know when and where these materials are actually relevant?

AUTHORS

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AFFILIATIONS

IFT _ Createch (A4)

RELATED LITERATURE

Andrady, A.L. (2011) Microplastics in the marine environment

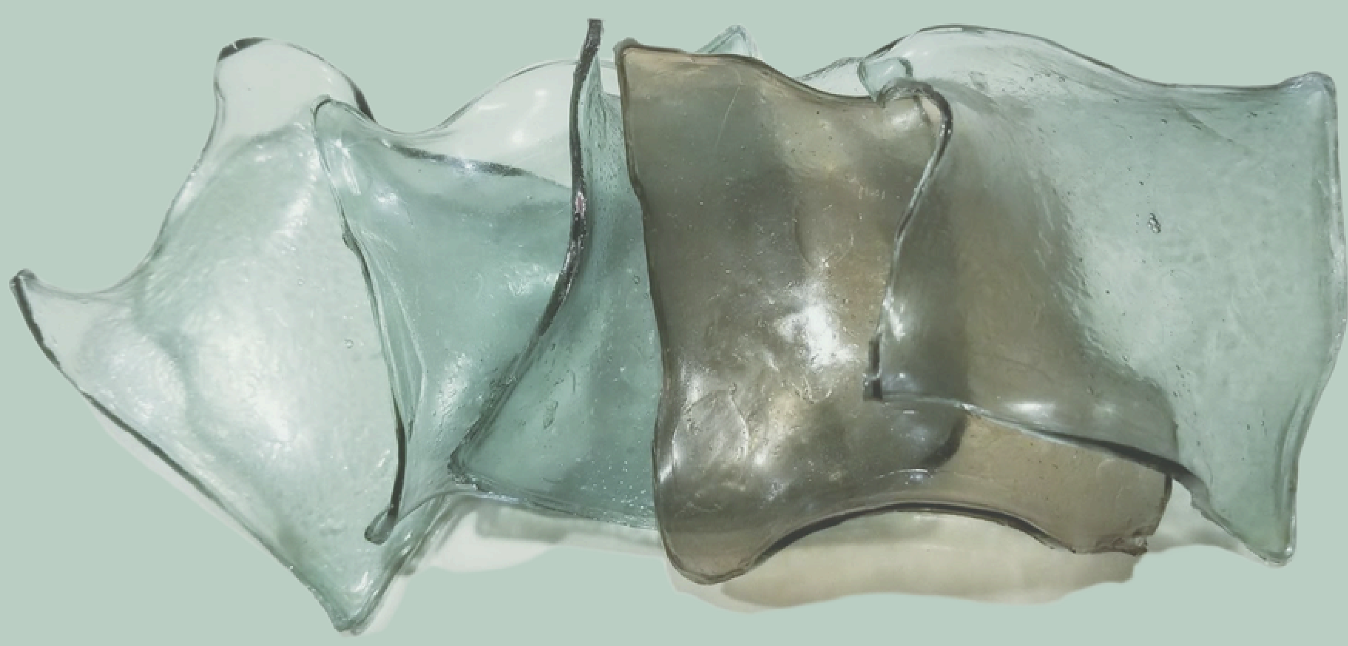
Taufik et al. (2020) The paradox between the environmental appeal of biodegradable plastics and their littering behavior

Sothornvit & Krochta (2001) Plasticizers in edible films and coatings ET

Vieira et al. (2011) Natural-based plasticizers and biopolymer films

Lee & Mooney (2012) Alginate: properties and biomedical applications

ASTM D1141-98 (révisé) : Standard Practice for the Preparation of Substitute Ocean Water



METHODOLOGY

I developed an experimental design space based on three key variables. First, the recipe, where the concentration of glycerine controls the material's flexibility. Second, the environment, comparing various seawaters to introduce different chemical conditions. Third, the exposure, including time, immersion versus evaporation, and light.

By combining these parameters, I produced a set of comparable samples, allowing me to observe and document how bioplastics transform under different conditions.

APPLICATION

Bioplastics were used in workshops with children designing sustainable cities, it worked (flexible, resistant). But when I said it was made from algae, they asked: "Can we throw it in the sea?"

This question shifted the project toward what happens after use.

OBJECTIVE

To understand what bioplastic films become after use. To explore how they behave in different environments. To reveal how their properties interact with context

RESULTS

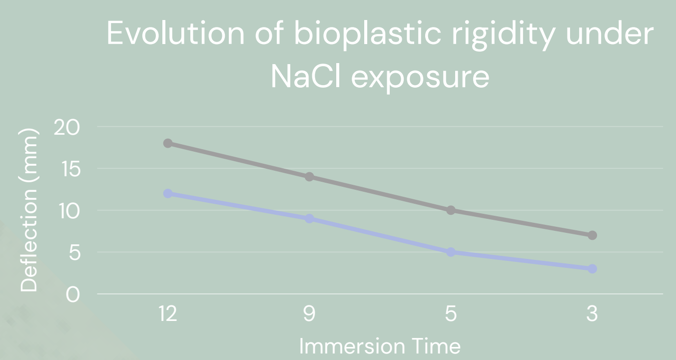
Bioplastics did not disappear. They transformed (deformation, opacity, rigidification, crystallization).

ANALYSIS

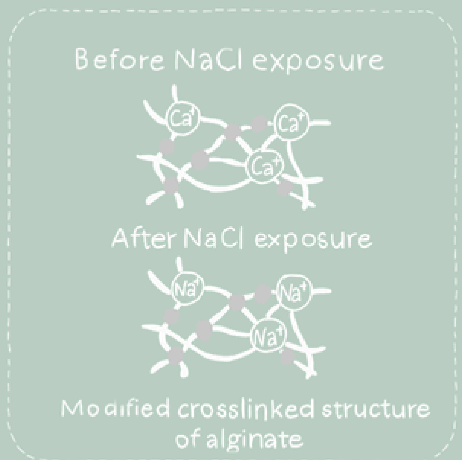
These transformations are driven by ionic interactions between the material and the environment.

In seawater, ions such as sodium and calcium interact with the alginate structure, modifying its internal bonds. This leads to changes in rigidity, texture, and structure.

The material is therefore not inert : it reacts with its environment. Bioplastics do not simply degrade; they are chemically transformed by the conditions they are exposed to.



The decrease in deflection over time indicates an increase in material rigidity.
This effect is observed for both formulations, with low glycerine samples stiffening more rapidly than high glycerine ones.



Ionic exchanges between Na^{+} and Ca^{2+} modify the crosslinked structure of alginate, leading to a reorganization of the material and an increase in rigidity.

LIMITS & CONCLUSION

This project shifts the focus from what bioplastics are to what they do over time. By confronting the material with a real environment, it reveals that its behavior is not fixed, but evolving and context-dependent. Rather than providing a definitive answer, it highlights a blind spot in material design: we design for use, but rarely for what comes after. Understanding materials through their transformations opens new ways of designing, not as stable objects, but as processes.